

Translated

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Solve RMQ (Range Minimum Query) by finding LCA (Lowest Common Ancestor)

Given an array $A[0..N-1]$. For each query of the form $[L, R]$ we want to find the minimum in the array A starting from position L and ending with position R . We will assume that the array A doesn't change in the process, i.e. this article describes a solution to the static RMQ problem

Here is a description of an asymptotically optimal solution. It stands apart from other solutions for the RMQ problem, since it is very different from them: it reduces the RMQ problem to the LCA problem, and then uses the [Farach-Colton and Bender algorithm](#), which reduces the LCA problem back to a specialized RMQ problem and solves that.

Algorithm

We construct a **Cartesian tree** from the array A . A Cartesian tree of an array A is a binary tree with the min-heap property (the value of parent node has to be smaller or equal than the value of its children) such that the in-order traversal of the tree visits the nodes in the same order as they are in the array A .

In other words, a Cartesian tree is a recursive data structure. The array A will be partitioned into 3 parts: the prefix of the array up to the minimum, the minimum, and the remaining suffix. The root of the tree will be a node corresponding to the minimum element of the array A , the left subtree will be the Cartesian tree of the prefix, and the right subtree will be a Cartesian tree of the suffix.

In the following image you can see one array of length 10 and the corresponding Cartesian tree.